

STAT 312: Midterm Exam 01 Announcement

Instructor: Dr. Md Mutasim Billah

EXAM 01 Topics

The following topics are covered on Exam 01.

- **Unit 02: Descriptive Statistics**

- Statistics, variables, populations, and samples.
- Numerical summaries: mean, median, mode, range, variance, standard deviation, percentiles, quartiles, and interquartile range (IQR).
- Visualizations: histograms, bar plots, dot plots, stem-and-leaf plots, and boxplots.

- **Unit 03: Probability**

- Sets, the empty set, subsets, sample spaces, and Venn diagrams.
- Intersection, union, complement, and mutually exclusive events.
- Conditional probability and independence.
- Multiplication rules, the law of total probability, the addition rule, the complement rule, De Morgan's laws, and Bayes' rule.
- Tree diagrams and probability tables.
- Drawing with or without replacement.
- Random variables.

- **Unit 04: Discrete Random Variables**

- Probability mass functions (PMFs) and cumulative distribution functions (CDFs).
- Population mean (expected value), variance, and standard deviation of a discrete random variable.
- Combinations and the binomial coefficient $\binom{n}{k}$.
- Binomial, geometric, and Poisson random variables.

- **Unit 05: Continuous Random Variables**

- Probability density functions (PDFs) and cumulative distribution functions (CDFs).
- Population mean (expected value) and variance of a continuous random variable.
- Uniform and exponential distributions.

- **Unit A: Joint Distributions**

- Discrete random variables: joint PMFs, marginal PMFs, conditional PMFs, and independence.
- Continuous random variables: joint PDFs, marginal PDFs, conditional PDFs, and independence.
- Covariance.

Exam 01 Preparation Guidelines

For each unit covered on Exam 01, review the following course resources thoroughly:

- Activities on the course website.
- Class lecture notes.
- Class handouts and your accompanying notes.
- Quizzes.
- Homework assignments.
- Required practice problems.

STAT 312: Midterm Exam 1 – Fall 2026

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Department of Mathematics, Applied Mathematics, and Statistics

Basic Statistics for Engineering and Science

Full Name: _____ Section: _____

Read each question carefully and answer all parts.

For multiple-choice and true/false/not-enough-information items, circle one answer.

For computational questions, show your setup, supporting work, and final answer, unless otherwise stated. Clearly box or circle your final answer.

You may use a physical calculator during the exam. Phone calculators are not permitted.

You may use one sheet of formula with writing on one side only. Writing on both sides is not permitted. You need to submit this with your answer paper.

Academic Integrity:

Students must complete the exam independently. Giving or receiving assistance and using unauthorized resources are prohibited. Permitted resources are limited to a physical calculator and one formula sheet with writing on one side only. ***Academic integrity is taken very seriously.*** Confirmed violations will result in a score of zero on the exam and may be reported to the Dean of Students in accordance with university policy.

Please read and sign the following honor pledge.

“On my honor, I have neither given nor received unauthorized aid on this examination.”

Signature: _____